

Research Seminar

Thesis Writing at the IMS

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Research Seminar $\leftrightarrow$ Research Module (reminder)

- **Research Seminar with Sabine**
 - no registration for any module or exam
 - tasks: obligatory but not graded
- **Research Module with your MSc thesis supervisor**
 - registration to be done together with thesis registration
 - tasks, all graded by thesis supervisor:
 - thesis proposal (start of thesis)
 - thesis talk+exam (end of thesis)

Steps Involved in Thesis Writing at the IMS

- ➊ Identify a research topic & find a supervisor (or vice versa)
- ➋ Register your thesis & write a proposal (intertwined)
- ➌ Perform your studies in interaction with your supervisor
- ➍ Write your thesis
- ➎ Hand in your thesis
- ➏ Present your thesis research
- ➐ Sit an oral exam related to your thesis research;
typically packaged with the thesis presentation (step 6)

General Information about Thesis Writing at the IMS

- Prüfungsordnung (Aug 3, 2017)
- Prüfungsordnung (Jul 30, 2024)
- Information about final theses at Uni Stuttgart
- Information about final theses at the IMS
 - topic + supervisor/examiner
 - registration (thesis and Research Module)
 - writing
 - handing in
- ILIAS materials regarding student theses at the IMS (topics, formalia, examples, grading)
- ILIAS materials regarding student writing at the IMS

IMS Thesis Topics

- IMS research groups
- Slides from our Research Seminar group presentations
- Student thesis topics at the IMS (via ILIAS)
- IMS-external theses (only in German)
 - IMS-external two cases: Uni-internal but not IMS & Uni-external
 - You need a scientific CL question!
 - You want a competent external supervisor!
 - You still need an IMS supervisor/examiner, and your exam regulations (and not others) apply to your thesis setup!

Contact potential supervisor(s)!

IMS Thesis and RM Registration

You must register your Master's thesis with the **thesis registration form** and the **Research Module registration form** at the **Examination Office**.

IMS prerequisites: **You need to have passed your "Auflagen" and "MethodsCL" (original or CL replacement) before starting the thesis!**

- 1 **Confirm that you reached the required number of 60 ECTS credits.**
- 2 Signature of examiner in field "confirmation of topic assignment". (Signature of head of the examination committee is not necessary.)
- 3 **Complete the RM registration form** (pre-filled by Karin Leonte).
- 4 Seal and photocopy both forms for the IMS at the secretary.
- 5 Confirmation of registration at the Examination Office.
- 6 The **thesis** has to be handed in **6 months after registration**.
- 7 The **RM tasks (proposal, talk, exam)** will be **set up with your supervisor**. (Note, again: If you have an IMS-external supervisor, you still need an IMS supervisor for the RM tasks.)

- **MSc thesis room:** MSc CL students can use an office in our building, if they want to. Ask your supervisor to sign you up!
 - There are only a limited number of places (6–7 for working with an own notebook, 5 desktops).
 - No more than 15 students are allowed on this list.
 - For all students on the list: first come first serve every day.
- **Computational arrangements:** servers, spaces, etc.

Talk to your supervisor about which servers to use, where to store your data, etc.

IMS Thesis Proposal

- Length: $\approx$ 6–10 pages
- Structure and content: to be discussed with supervisor/examiner
(roughly: motivation; related work; material and methods; expected contributions/results; bibliography)
- Supervision effort:
 - ① discussion of structure+content between student and supervisor
 - ② comments from supervisor on first draft of proposal
 - ③ second proposal version is graded
- Example proposals available on ILIAS

- A `TEX` minimal example template is available.
- Language:
 - English
 - If the supervisor/examiner agrees, you can choose a different language. In this case, you need to include an English abstract.
- Title page: name und address of the IMS, title of the thesis, name of author, two examiners, supervisor(s), date of beginning and end
- Declaration that the thesis was completely done by yourself.

IMS Thesis Writing: Use of AI (1)

Use of AI by Students at the University of Stuttgart

- Use of AI is permitted

I am allowed to use AI systems at the University of Stuttgart within the framework of these principles. My use of AI should support – not replace – my personal engagement with course content, my independent thinking, and the development of my academic skills.

- Decision on the use of AI in examinations

The extent to which the use of AI systems is permitted or encouraged in examination performances is determined by the respective examiners. In case of doubt, I ask the examiners for clarification.

- Responsibility for input

I am aware that I am responsible for all content I enter into an AI system. I do not enter any copyrighted material (text, images, etc.) into AI systems that store such content or use it for training purposes. I do not enter any confidential, internal, or personal data into AI systems unless explicitly authorized by a regulation of the University.

IMS Thesis Writing: Use of AI (2)

- Responsibility for output

I am responsible for the use of any output generated by AI. I am aware that AI systems can produce incorrect or discriminatory content and may hallucinate information. Before using any AI-generated output, I carefully review it for accuracy, potential similarities with existing works (unintentional plagiarism), and compliance with legal and ethical standards.

- Transparency and fairness in AI use

In consultation with the examiner, I clearly and transparently document and indicate where, how, and which AI systems I have used in order to ensure traceability. By adhering to these principles, I contribute to fairness in academic study and assessment.

- University-operated AI systems and fair use

I preferentially use GDPR-compliant AI systems provided by the University. When using university-operated systems, I act fairly and in a resource-conscious manner to ensure equitable access for all users.

- AI competencies

To strengthen my AI skills, I can take advantage of the University's training and learning opportunities

IMS Thesis Timeline and Submission

- Time: 6 months
- Hand in the thesis at the secretary's office (Karin Leonte).
- Hand in signed submission form and licensing agreement (without supervisor's signature).
Translation available on demand from Stefanie Anstein.
- Number of hard copies (identical casebound versions, adhesive if possible, all with titlepage and signed declaration, no scan or copy):
 - 1 for each examiner (2 examiners for Master's theses)
 - 1 for the archive (via the secretary)
 - new (from PO 2024): only 1 copy altogether
- Hand in a copy of the title page and an electronic version (CD only with thesis text in PDF and abstract in TXT format).
- Note: You do not need to be enrolled to submit your thesis (if it is your last module), you only need to be enrolled for registering it. (But: Enrollment is still needed to extend your visa and/or be able to work.)
- Credits: 30 ECTS (900 hours)

- Grading criteria available on ILIAS
 - IMS introduced standards for criteria
 - available on ILIAS: Student theses at the IMS → Thesis (and Report) Grading Criteria
 - weighting of the grading criteria is up to the supervisor/examiner, and also depends on the thesis type
 - your supervisor/examiner should introduce+discuss them with you
- Example theses available on ILIAS

IMS Thesis Supervision & Evaluation: Criteria Overview

- **Organisation**
 - self-organisation
 - commitment
 - interaction with the supervisor
- **Scientific work**
 - clear definition of research hypotheses and goals
 - coherence
 - completeness
 - correctness
 - originality
 - sustainability
- **Written document**
 - meaningful structure
 - coherence
 - quality of writing
 - quality of the presentation
 - metalevel
 - references

IMS Thesis Presentation & Exam

- Presentation:
 - takes place at the end of the thesis (before or after handing in)
 - length: 25–30 minutes
 - public discussion (IMS research group/staff members; family/friends)
- Oral exam:
 - typically directly following the presentation
 - content: supervisor (who is also typically the examiner) selects exam material and sends it to the student (ca. 2–3 weeks before the exam)
 - length: 20–30 minutes
 - not public; minimally one examiner and somebody to take notes

① Master's Thesis (30 ECTS)

② Research Module (9 ECTS)

- $\frac{1}{3}$ proposal
- $\frac{1}{3}$ presentation
- $\frac{1}{3}$ oral exam